

27.

Ans: A

energy of photon = energy of photoelectron + work function

$$hf = eV_s + \phi$$

$$V_s = \frac{h}{e}f - \frac{\phi}{e}$$

Since ϕ decreases, the vertical intercept decreases, resulting in a decrease in the horizontal intercept and a leftward shift of the graph.

Since h and e are constants, the gradient is unchanged.



